

# Candidate Intelligence Engine - Setup Instructions

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Welcome to the Candidate Intelligence Engine! This document provides detailed, step-by-step instructions on how to set up, run, and evaluate this project on your local machine. It is designed to be easily understandable, even if you do not have a technical background.

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## 1. Environment Setup

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- **Install Python:** Ensure you have Python 3.10 or higher installed. You can verify this by running `python --version` in your terminal.
  - **Open Terminal:** Open your Command Prompt (CMD) or PowerShell and navigate to the root directory of this project (`efp/`).
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## 2. Install Dependencies

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Install the required Python libraries using pip: `pip install -r requirements.txt`

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## 3. Run the Application

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This project operates without complex Docker or Kubernetes configurations to ensure local portability. You only need a single command to launch both the backend processing API and the dynamic UI dashboard.

```
python -m uvicorn candidate_transformer.api.app:create_app --factory
```

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## 4. Evaluate the Solution

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Once the server is running, open your web browser and navigate to:

<http://127.0.0.1:8000>

## Dashboard Workflows

1. **Upload Data:** You will be presented with a modern upload interface. You can drag-and-drop a candidate's Resume PDF into the upload area.
  2. **Data Extraction:** Click **Transform Data**. The intelligent backend pipeline will automatically parse the PDF, extract invisible hyperlink annotations (for portfolio URLs), deduplicate emails/phones, and resolve conflicting information.
  3. **View Results:** The output bar will dynamically appear.
    - **Interactive View:** An immersive graphical timeline of the candidate's Experience, Projects, Education, and Skills.
    - **Textual (JSON) View:** A syntax-highlighted display of the *Canonical JSON Profile*, perfectly matching the requested schema. Use this view to verify the structural integrity of the output.
    - **Quick Actions:** Easily **Copy** the JSON directly to your clipboard or **Download** the raw **.json** file for the canonical record!
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## 5. How It Works (For Non-Technical Users)

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Behind the scenes, the system acts like an ultra-fast recruiter:

- **Reading:** It reads the raw PDF (or CSV) and intelligently highlights names, emails, phone numbers, and past work experience. It even looks "under the hood" of the PDF to find hidden website links.
  - **Merging:** If it finds two different files for the same person, it seamlessly combines them like a puzzle, ensuring no duplicate information is saved.
  - **Standardizing:** Finally, it formats all this chaotic data into a strict, organized standard (JSON) that modern HR applications require to operate.
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## 6. System Architecture Breakdown

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For technical reviewers, the codebase is organized as follows:

- **src/api/**

- Serves the FastAPI web server, routes incoming uploads, and hosts the static dashboard (HTML/JS/CSS).

- **src/extraction/**

- The heart of the extraction intelligence. Uses robust regex, positional tracking, and advanced PDF annotation scraping to pull unstructured text and hidden URLs.

- **src/merge/**

- The resolution engine. Intelligently cross-references and merges overlapping attributes across multiple file sources into one cohesive canonical source of truth.

- **src/projection/**

- The final transformation step. Carefully aligns the merged data structure into the strict, canonical JSON schema required by downstream systems.

Enjoy exploring the Candidate Intelligence Engine!